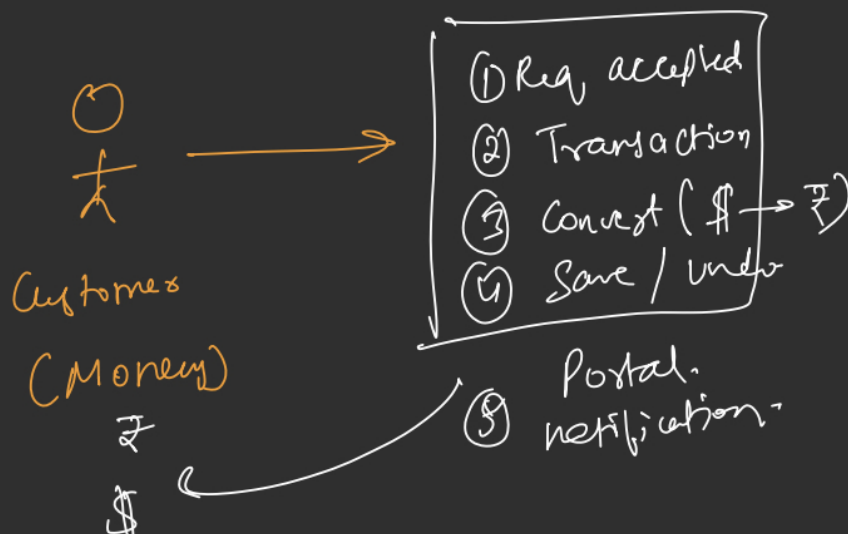


Forex Money Exchange

Requirements:

- A customer can exchange currency.
- We should have all the latest exchange rate.
- Request will be fulfilled FIFO.
- VIP customers will be given priority.
- Transaction save / undo.
- History show.
- Notification to Observers.
(SMS, File, printer)
- Few multiple algo 



→ Strategy Design pattern.

→ Observer pattern ^{or}

→ Singleton.

DS

↳ Queue

↳ PG

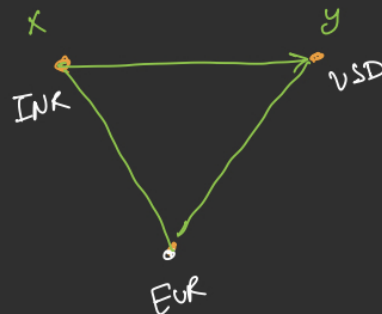
↳ stack

↳ map

} List }

∞ profit

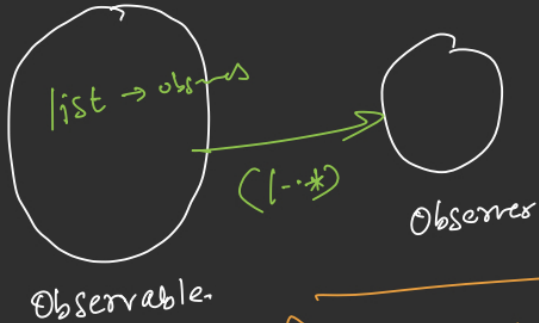
Best Exchange Rate



Observer Design Pattern

NSF/BSE

Zomato → . .



Pull-based (polling)
↓
Push-based

```
Class Observable {  
    vector<Observer>  
        observers;  
    Change() { }  
    for (o : observer) {  
        o → update();  
    }  
}
```

```
Class Observer {  
    update();  
}
```

```
Class Ob1 : Observer {  
    update() { }  
}
```

```
Class Ob2 : Observer {  
    update() { }  
}
```

→ Notification

Actors

↳ Customer

↳ Money (value, currency)

↳ Currency (id, name)
code

↳ Exchange Rate

↳ Transaction

↳ Fees

Orchestration

Forex Exchange Service

Principle of least
Knowledge

